

Machine Learning Principles

Class 18: November 10

Bayesian Net II

Queries: conditional independence & posterior probability (inference)

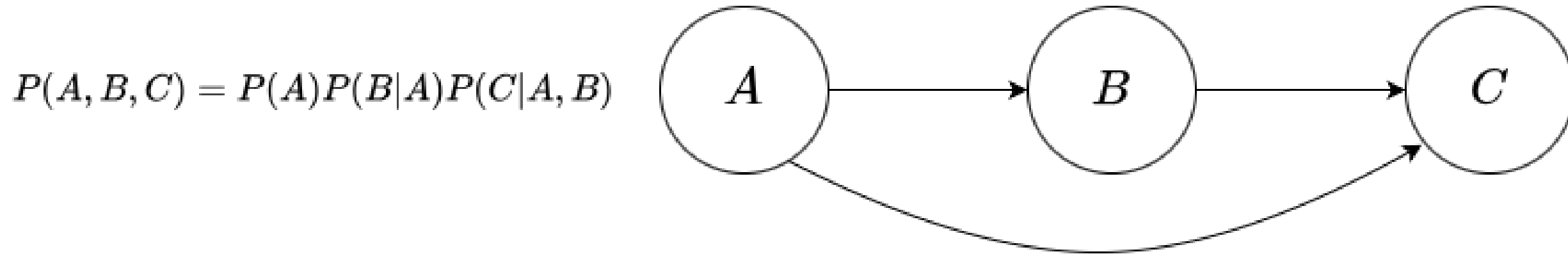
Learning: EM algorithm

Instructor: Diana Kim

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***Bayesian Net

- A joint density
can be represented by the product of conditional factors
by a **Directed Acyclic Graph (DAG)**: Bayesian Net

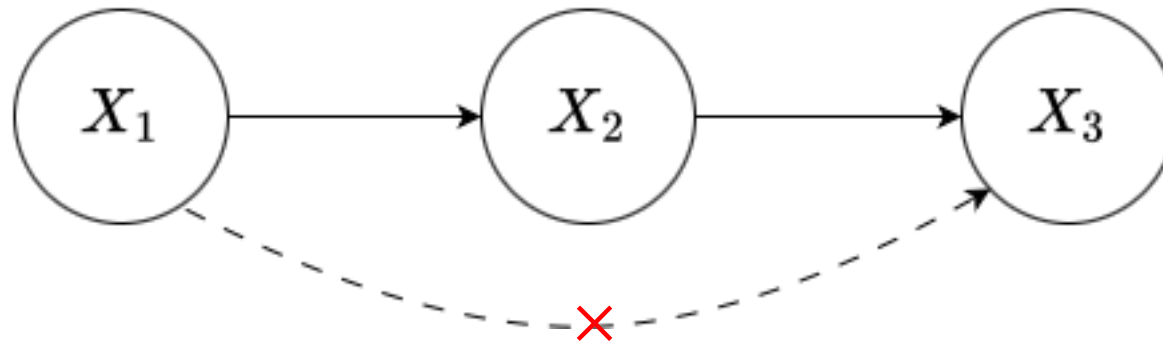


Q: learning a Bayesian Net: learning what?

(1) connectivity (edges) (2) conditional densities for each vertex

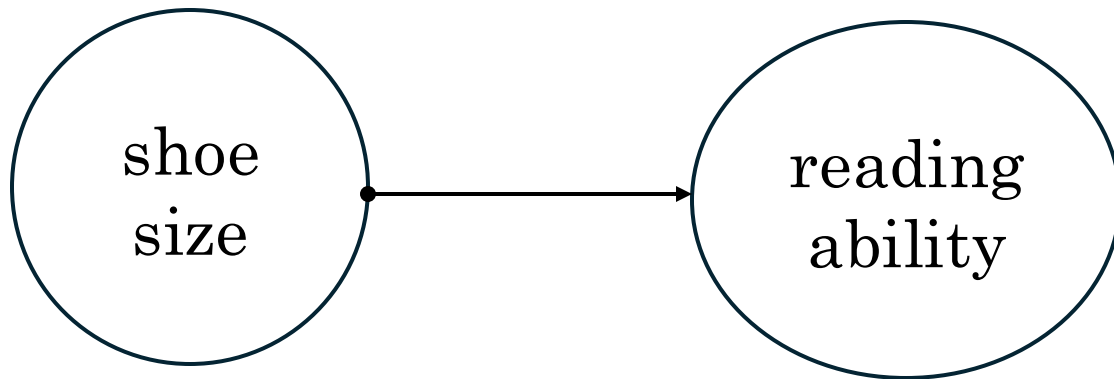
***Bayesian Net Representation (missing edges encodes CI)

$$P(X_1) P(X_2 | X_1) P(X_3 | X_2)$$



***Bayesian Net Representation (causal and effect?)

Q: arrows A to B in a Bayesian network implies
a causal (A) and effect (B) relationship? they encode correlation and
not cause & effect



***Bayesian Net (what we can do with BN?**)

- [1] determine if a set of random variables are conditionally independent given some observation?
sometimes it is not decidable.
- [2] cause and effect $P(B|C, A) = P(B|C) \rightarrow A$ is not the cause of B
- [3] querying the posterior prob like $P(X, Y|z)$?

Today's Lecture

1. Bayes Net Application.

(1) Conditional Independency

- elementary blocks for Bayesian network (three nodes and two edges)
chain structure, tent (fork) structure, and v-structure

(2) Inference: computing the posterior probability

- variable elimination
- Markov Chain Monte Carlo (MCMC) simulation (Gibbs Sampling)

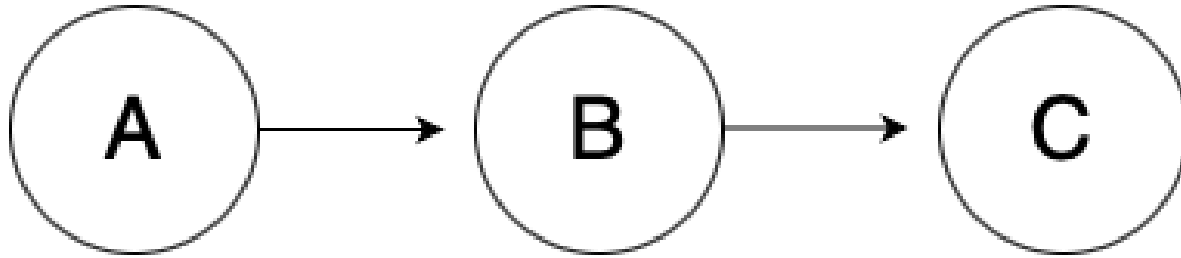
2. Learning a Bayesian Net (learning parameters : CPT)

- a network structure is known and data is fully observed
- a network structure is known and data is partially observed
- **E**xpectation **M**aximization (EM) algorithm (partial observation)

- Elementary Blocks of Bayesian Net (three vertexes and two edges)
 - chain structure
 - tent structure
 - V-structure

[1] Elementary Blocks (chain structure)

- **Chain Structure**: the middle node of chain breaks into two.



B mediator screen off information about A from C: $A \perp C \mid B$

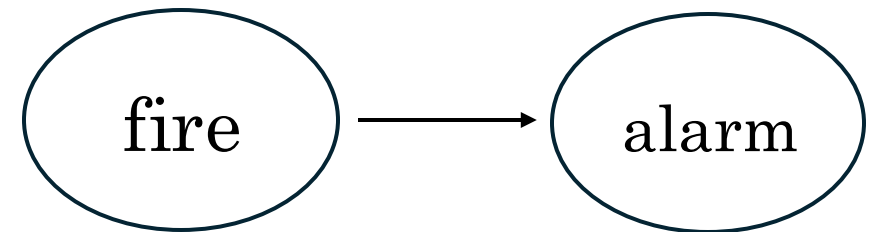
[2] Elementary Blocks (chain structure example)

- **Chain Structure:** the middle node of chain breaks into two.



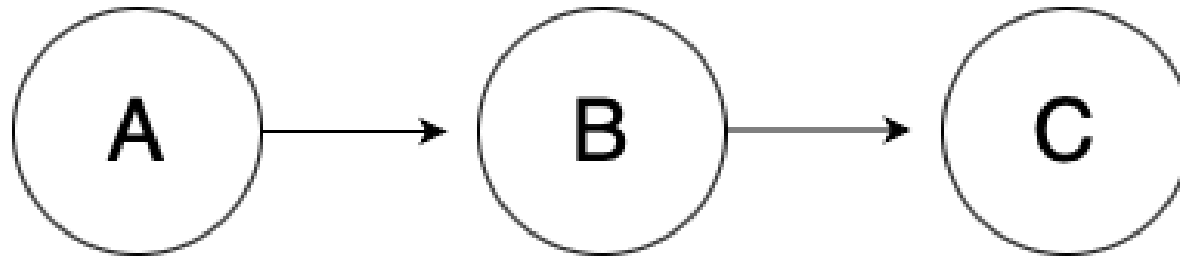
the fire by itself does not set off an alarm.

Q: is it possible? fire is a cause for the alarm?



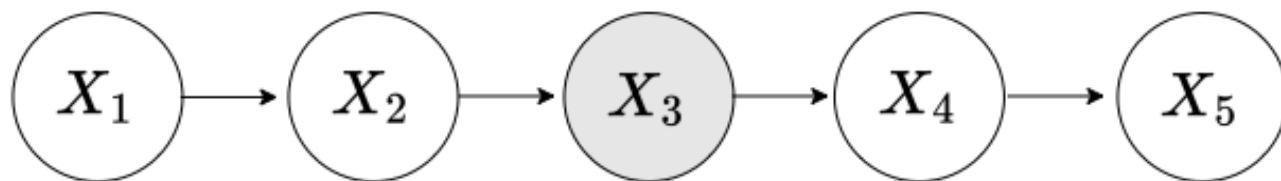
[3] Elementary Blocks (chain structure)

Q: a possible conclusion about causation from the chain structure ?



[4] Elementary Blocks (extended chain structure)

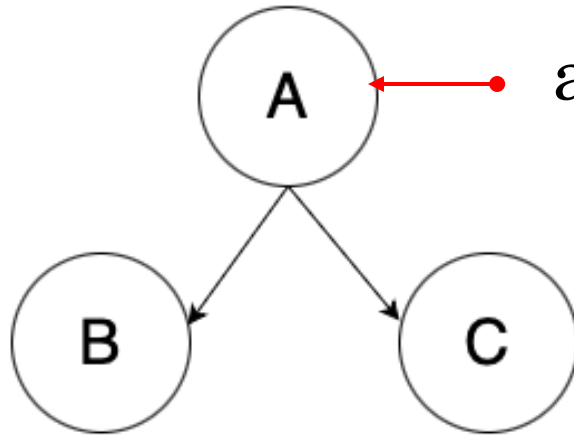
- extended chain structure: the middle node of chain breaks into two.



$$\begin{aligned} P(X_1, X_5 | X_3) &= \frac{P(X_1, X_3, X_5)}{P(X_3)} \\ &= \frac{\sum_{X_2, X_4} P(X_1, X_2, \dots, X_5)}{P(X_3)} \\ &= \frac{\sum_{X_2, X_4} P(X_1, X_2, X_3) P(X_4, X_5 | X_1, X_2, X_3)}{P(X_3)} \\ &= \sum_{X_2, X_4} P(X_1, X_2 | X_3) P(X_4, X_5 | X_3) \\ &= P(X_1 | X_3) P(X_5 | X_3) \end{aligned}$$

[5] Elementary Blocks (tent structure)

- tent (fork) structure: root node separates its children

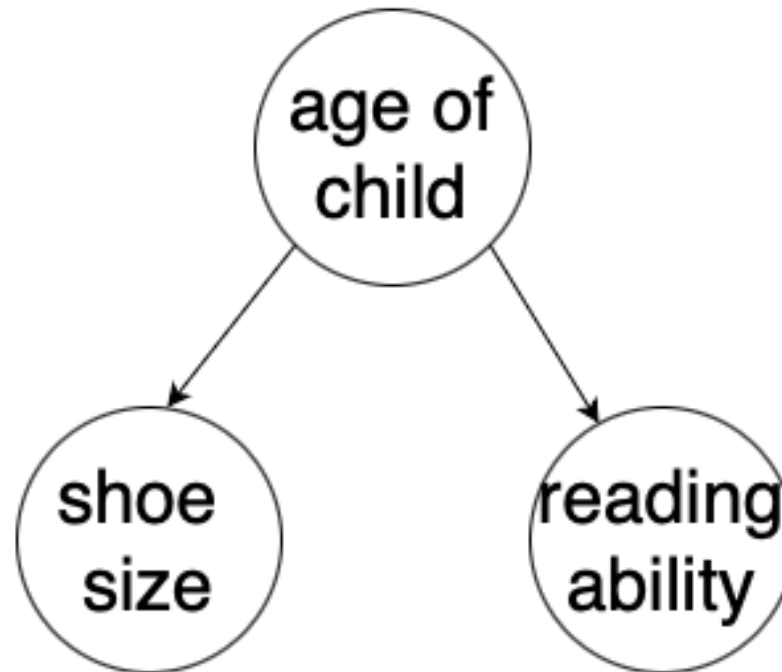


- the cofounder A makes B and C are statistically correlated.
- B and C are correlated without A
- $B \perp C \mid A$

$$\begin{aligned} P[B, C|A] &= \frac{P[ABC]}{P[A]} \\ &= \frac{P[AB]P[C|A]}{P[A]} \\ &= P[B|A] \cdot P[C|A] \end{aligned}$$

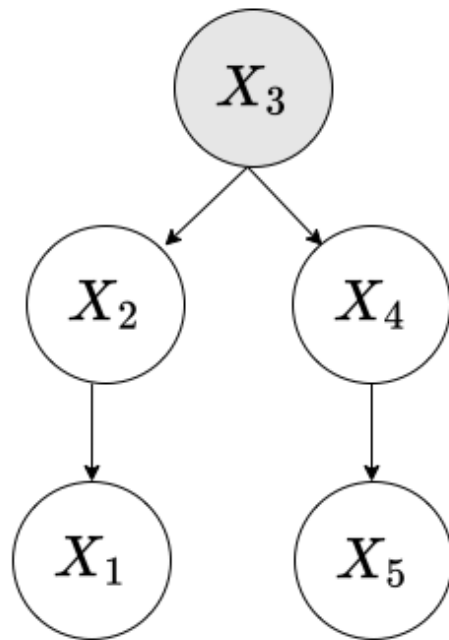
[6] Elementary Blocks (tent structure example)

- tent (Fork) structure: root node separates its children



[7] Elementary Blocks (extended tent structure example)

- extended tent structure: root node separates its children

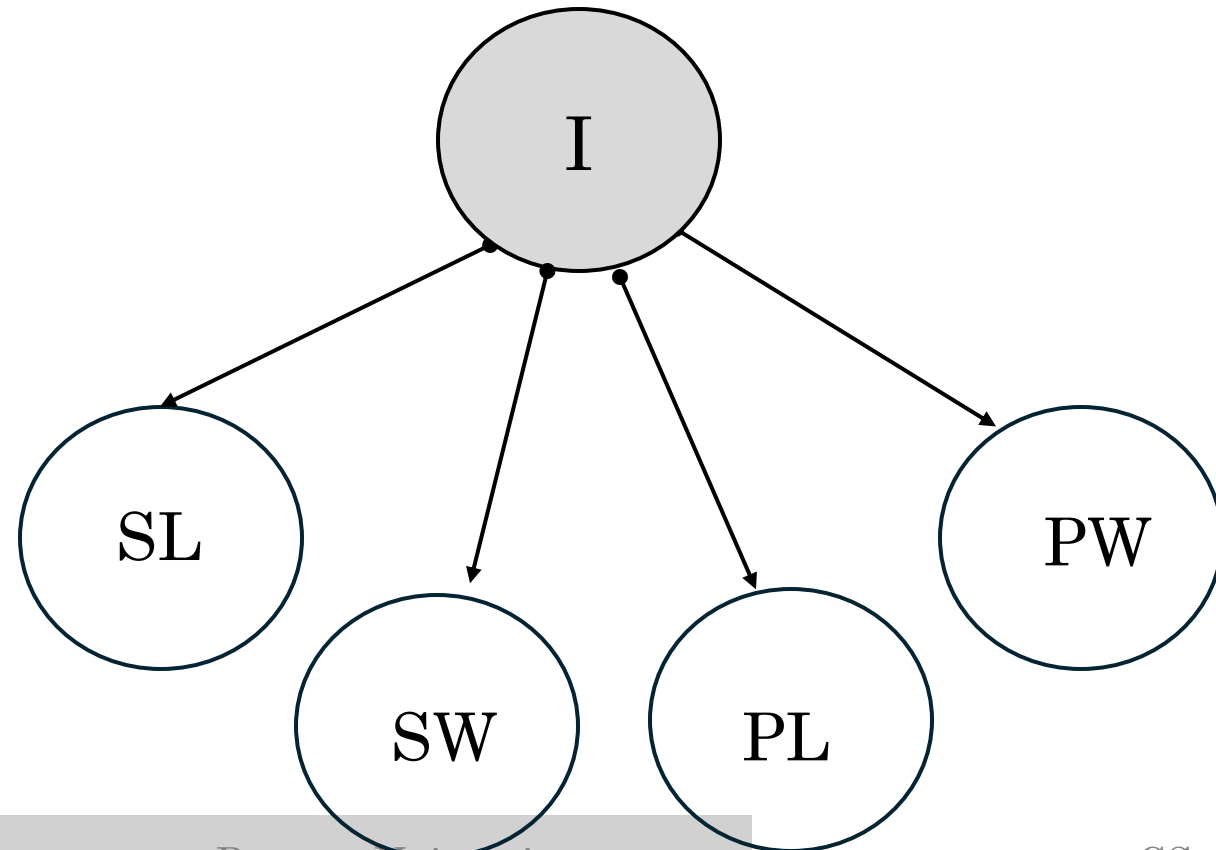


$$\begin{aligned} P(X_1, X_5 | X_3) &= \frac{P(X_1, X_3, X_5)}{P(X_3)} \\ &= \frac{\sum_{X_2, X_4} P(X_1, X_2, \dots, X_5)}{P(X_3)} \\ &= \frac{\sum_{X_2, X_4} P(X_1, X_2, X_3) P(X_4, X_5 | X_1, X_2, X_3)}{P(X_3)} \\ &= \sum_{X_2, X_4} P(X_1, X_2 | X_3) P(X_4, X_5 | X_3) \\ &= P(X_1 | X_3) P(X_5 | X_3) \end{aligned}$$

[8] Elementary Blocks (tent structure example)

[example of tent structure]

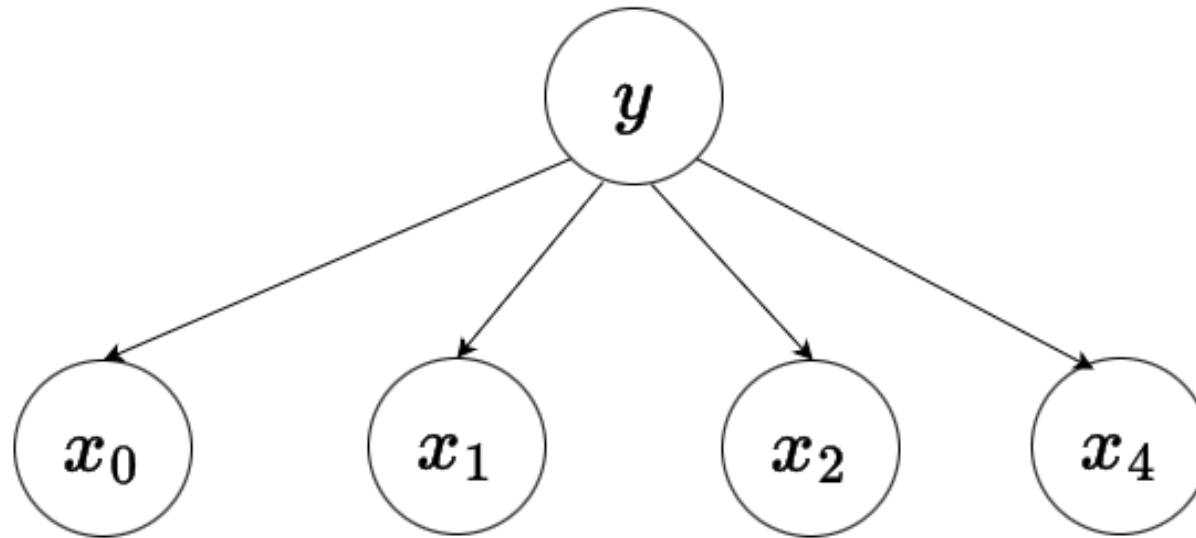
In the hw1, the feature of SL/ SW/ PL/ PW were conditionally independent given Iris information.



- SL: sepal length
 - SW: sepal width
 - PL: pedal length
 - PW: pedal width
-
- when Iris information observed, SL/ SW/ PL/ PW becomes independent.

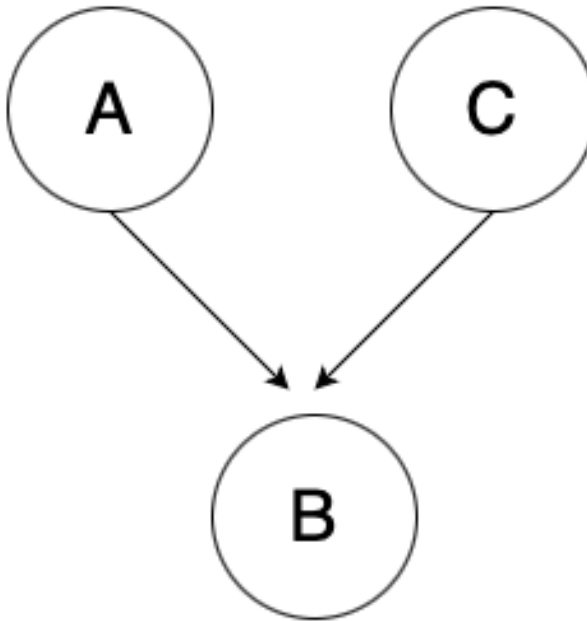
[9] Elementary Blocks (tent structure of naïve Bayes)

Conditional Independence Assumption for Naïve Bayes

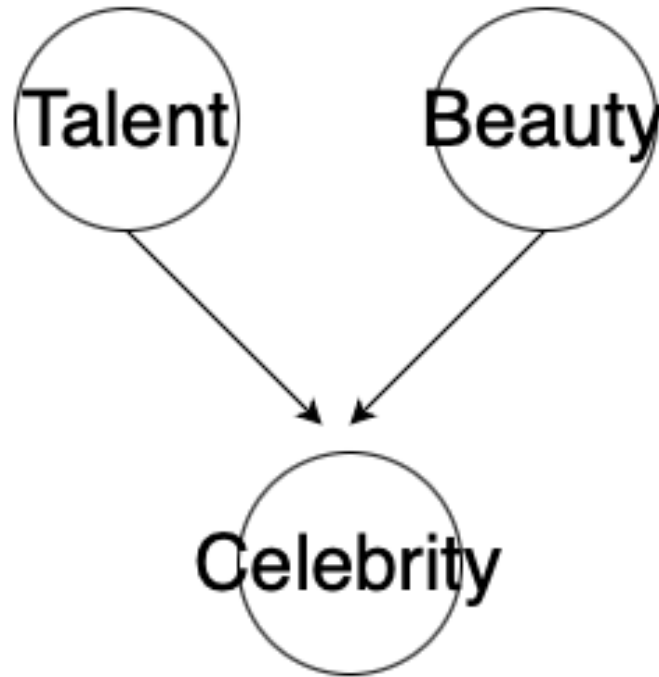


[10] Elementary Blocks (V structure)

- *V*-Structure (collider):
conditioning on a common child/ (or a descendant of the child) or at the bottom of a v-structure makes its parents become dependent.



[11] Elementary Blocks (V structure example)



both talent and beauty contribute to an actor's success, but beauty and talent are completely uncorrelated. If a celebrity is a particularly good actor that explains away the person's success, the person does not need to be more beautiful than the average person.

- D-separation (a criterion to decide a set of A and B independent given e)
 $[A \perp B | e]$

[1] D separation (definition)

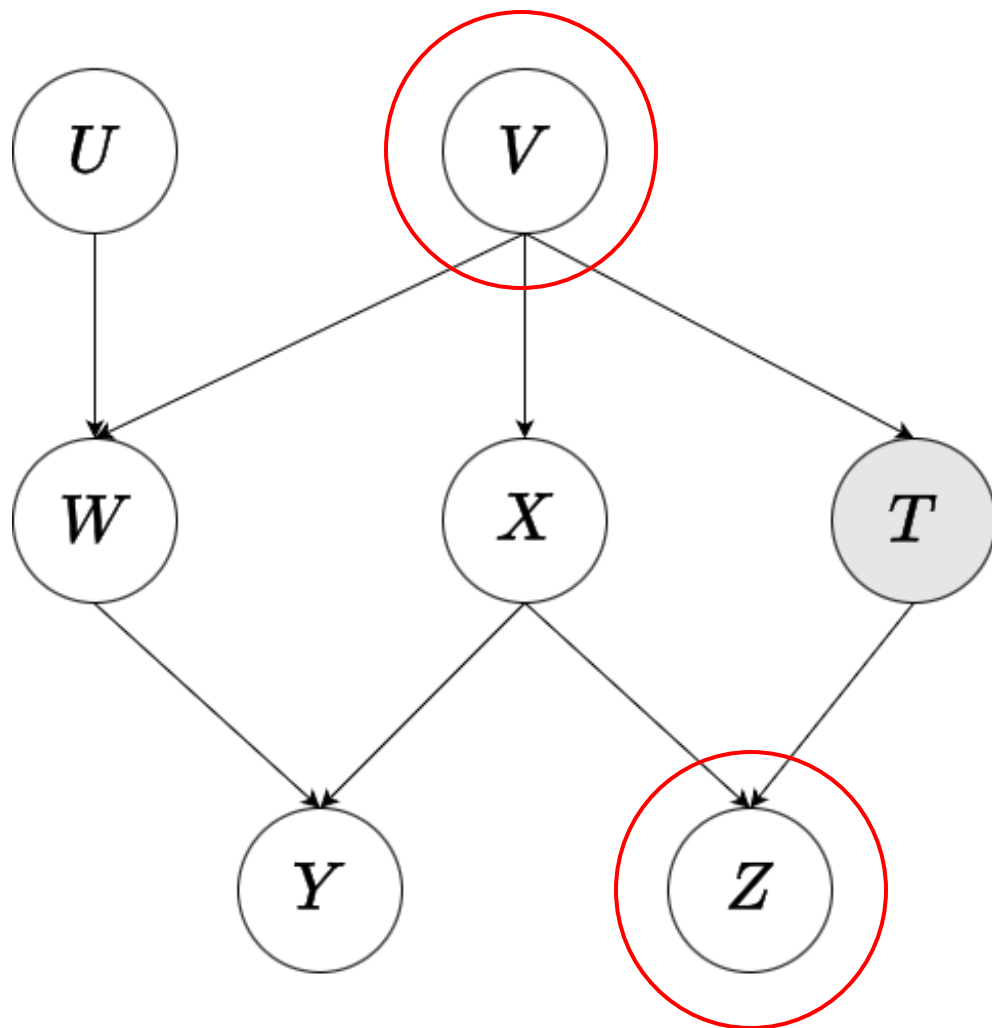
[D separation between two nodes]

The node A is **d-separated** from B by a set of observation e
 $\Leftrightarrow$ each (all) undirected path from A to B is d-separated.

$$\Leftrightarrow [A \perp B \mid e]$$

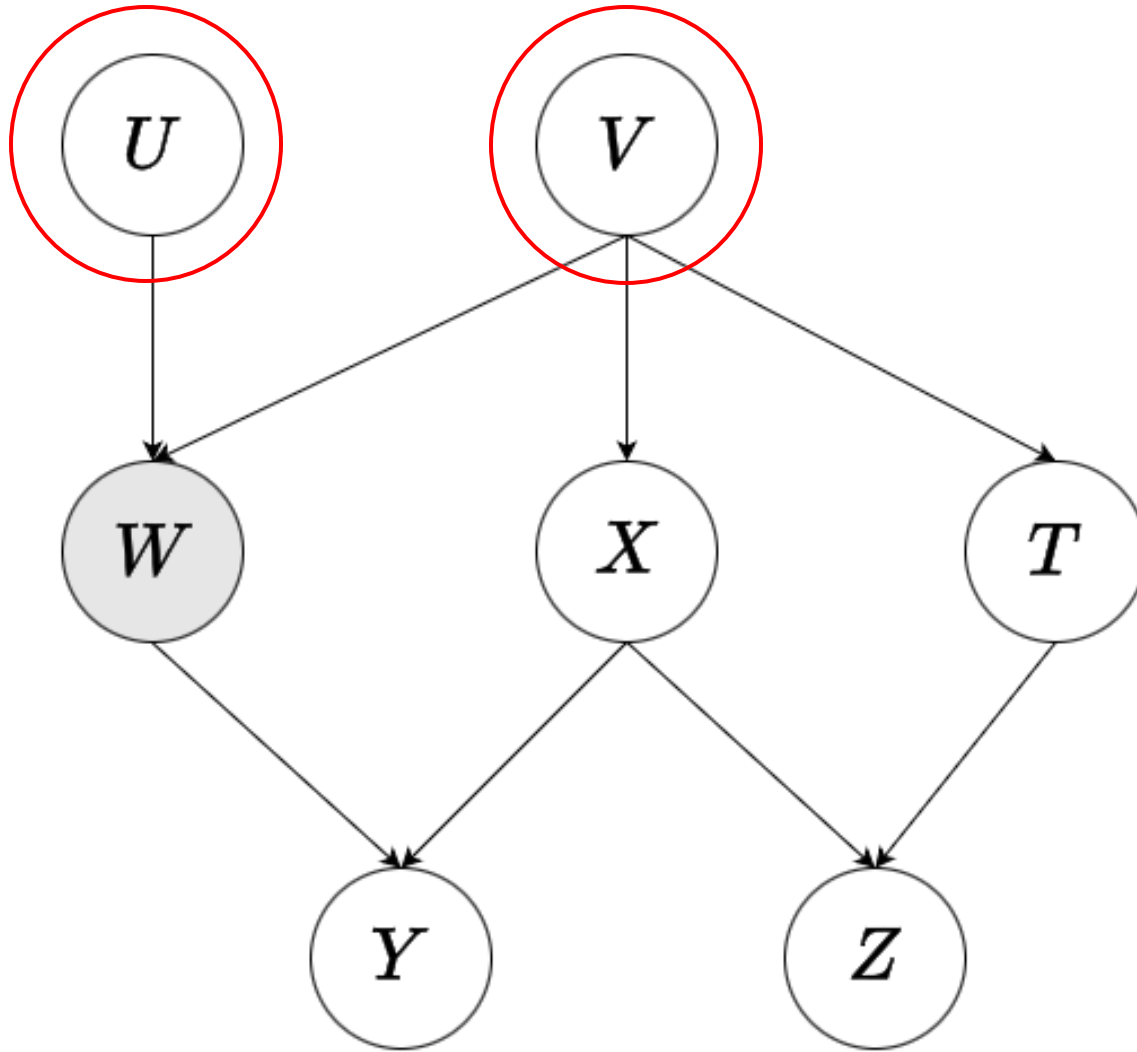
Q: What if we find a path between A and B that is not d-separated?

[2] D separation (example1)



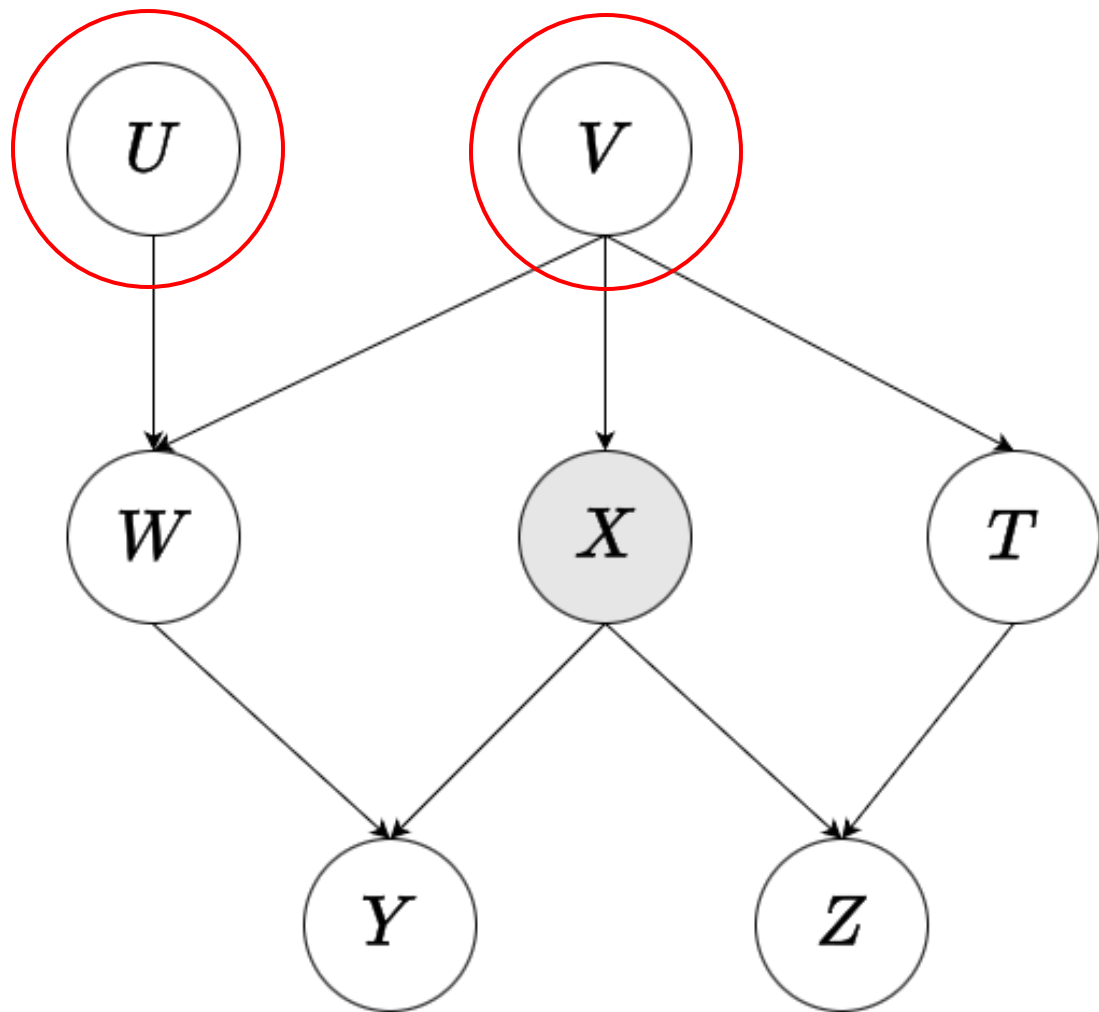
$Q: [V \perp Z|T]?$

[3] D separation (example2 **recitation)



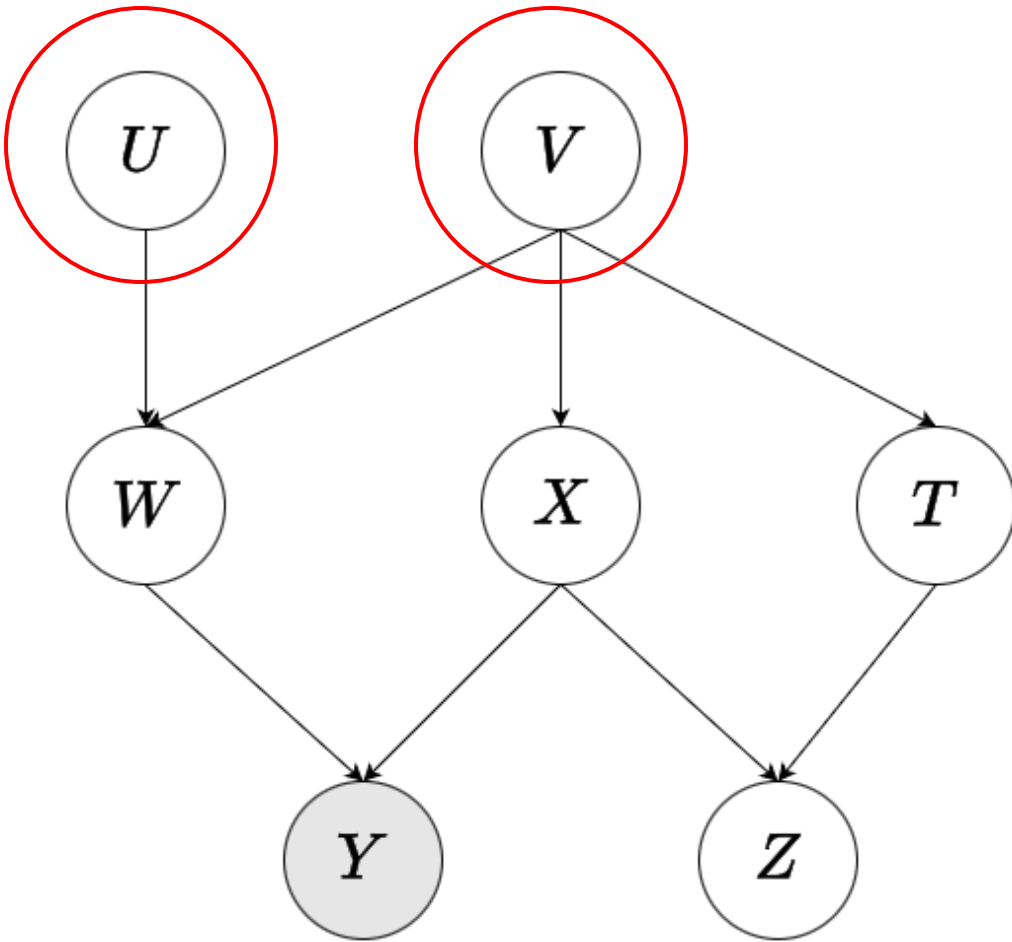
$Q: [U \perp V | W]?$

[4] D separation (example3 **recitation)



$Q: [U \perp V | X]?$

[5] D separation (example4)



$Q: [U \perp V | Y]?$

[6] D separation (why it matters?)

How D-separation matter?

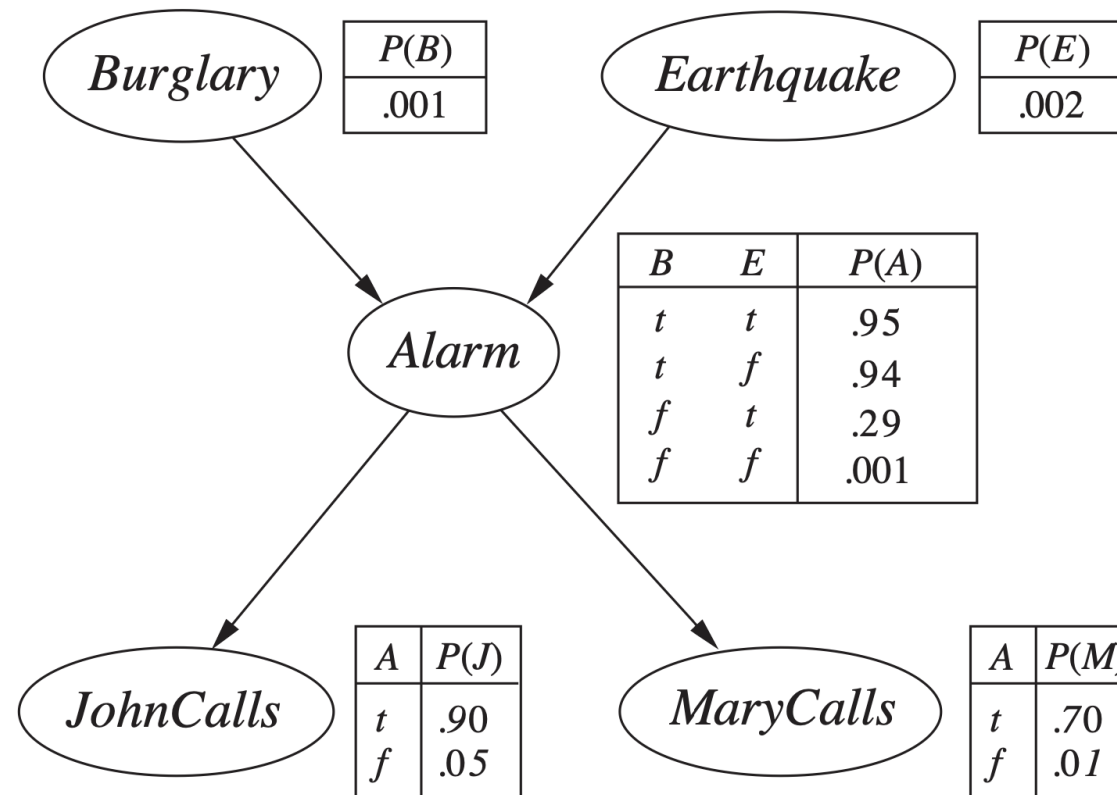
How $A \perp B \mid e$ matter?

What conclusion can be derived from it about causal relationship ?

[7] D separation (why it matters?)

In the previous slide,
Depending on the order of variable introduction,
the same density can be represented by various ways.

[8] D separation (why it matters?)



How the structure will be changed as the order of
MaryCalls $\rightarrow$ JohnCalls $\rightarrow$ Alarm $\rightarrow$ Burglary $\rightarrow$ Earthquake?

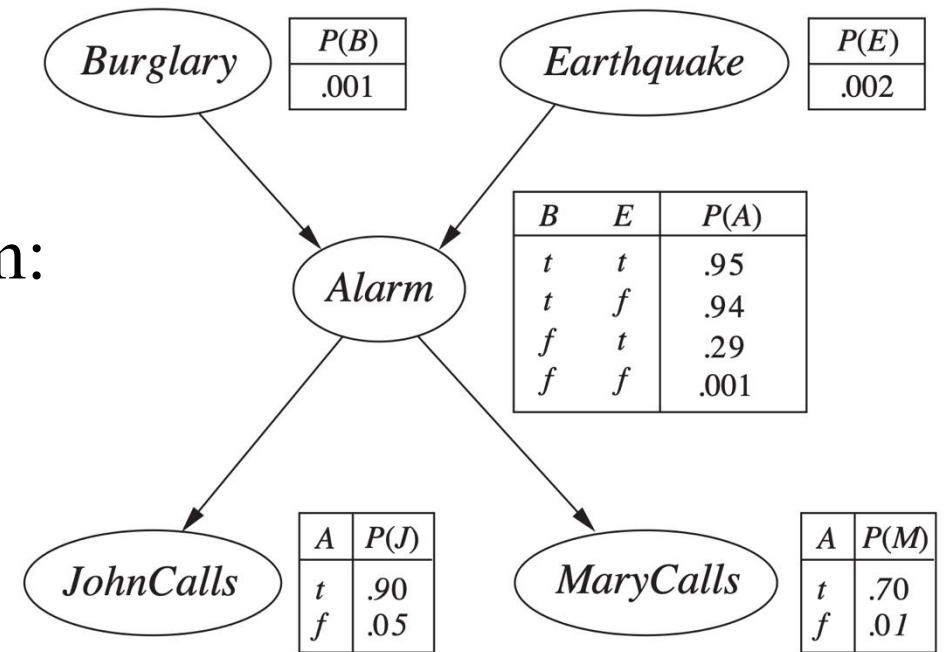
[10] D separation (why it matters?)

Bayesian network based on new order

:MaryCalls → JohnCalls → Alarm → Burglary → Earthquake?

For the new structure, the following conditional independence are derived from the original structure.

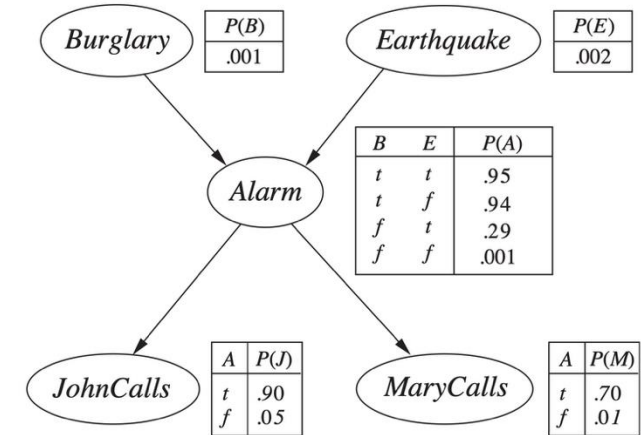
- Burglary $\perp$ MaryCalls and JohnCalls | Alarm:
- Earthquake $\perp$ MaryCalls, JohnCalls | Alarm
- Earthquake $\not\perp$ Burglary | Alarm



[11] D separation (why it matters?)

Bayesian network based on new order

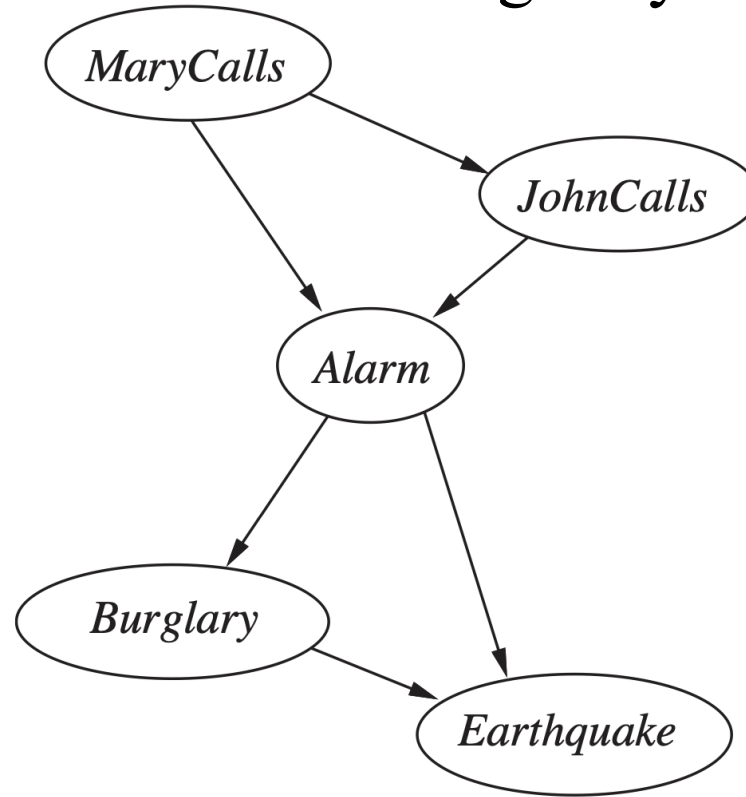
:MaryCalls $\rightarrow$ JohnCalls $\rightarrow$ Alarm $\rightarrow$ Burglary $\rightarrow$ Earthquake?



[9] D separation (why it matters?)

Bayesian network based on new order

:MaryCalls $\rightarrow$ JohnCalls $\rightarrow$ Alarm $\rightarrow$ Burglary $\rightarrow$ Earthquake?



From Figure 14.3 in Artificial Intelligence: A Modern Approach by Russell

- Inference: Computing the Posterior Probability
how to compute $P[\text{Burglary} \mid \text{MaryCalls} + \& \text{JohnCalls} +]?$

**handling the posterior queries is the main reason of Bayesian net is invented for.
(reduce the computational complexity in marginalization)

❖ Exact Inference

[1] Inference (computing posterior prob)

The general format of query

$$P(X|e) = \frac{P(X, e)}{P(e)} = \frac{\sum_H P(X, H, e)}{P(e)} = \alpha \sum_H P(X, H, e)$$

- X query variable
- H non-query variable
- e evidence variable

Q: Bayesian net provides a joint density,
so any type of query can be answered?

[2] Inference (computing posterior prob)

Given $P(X_1, X_2, X_3)$

$$P(X_1 | X_2 = x_2)? = \frac{P(X_1, X_2 = x_2)}{P(X_2 = x_2)} = \alpha \sum_{X_3} P(X_1, X_2 = x_2, X_3)?$$

**Inference (computing posterior prob)

- without CI

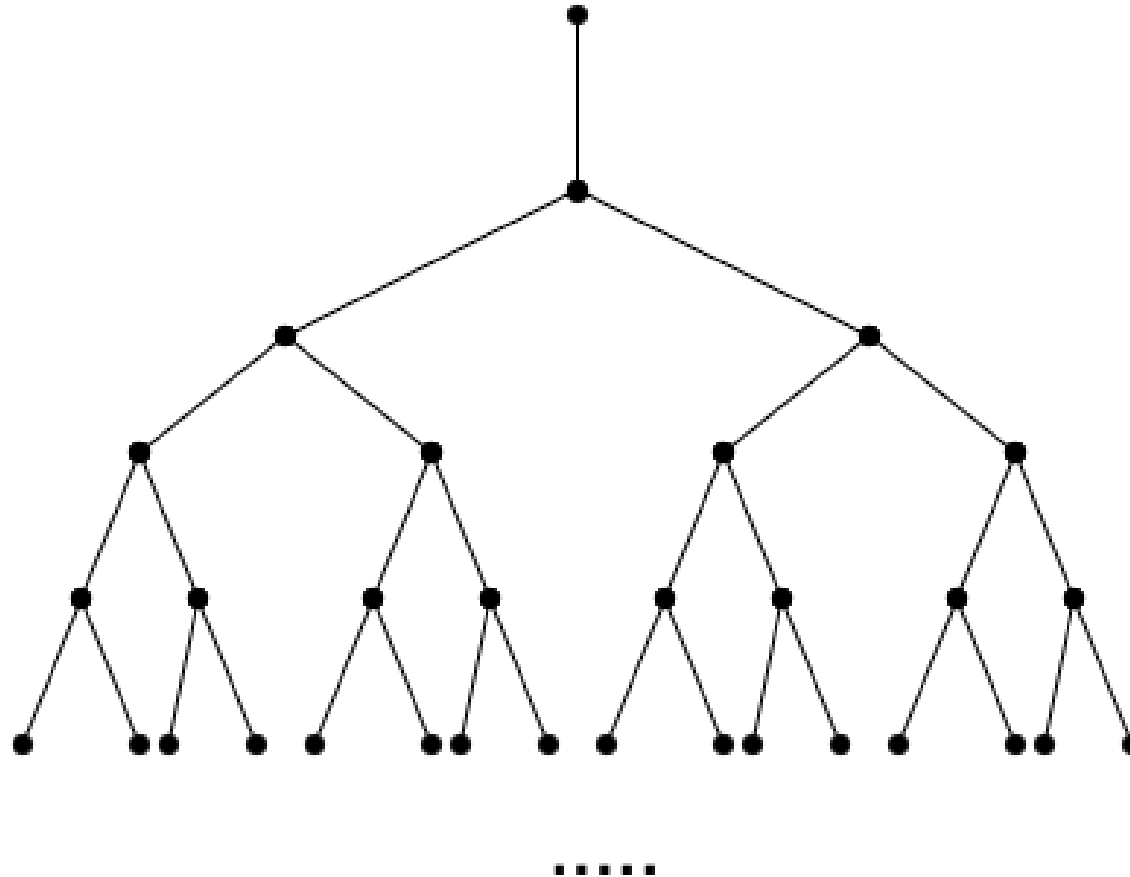
$$P(X_{36}) = \sum_{X_1} \sum_{X_2} \dots \sum_{X_{35}} \sum_{X_{37}} \dots \sum_{X_{1000}} P(X_1, X_2, \dots, X_{35}, X_{36}, X_{37}, \dots, X_{1000})$$

- with CI

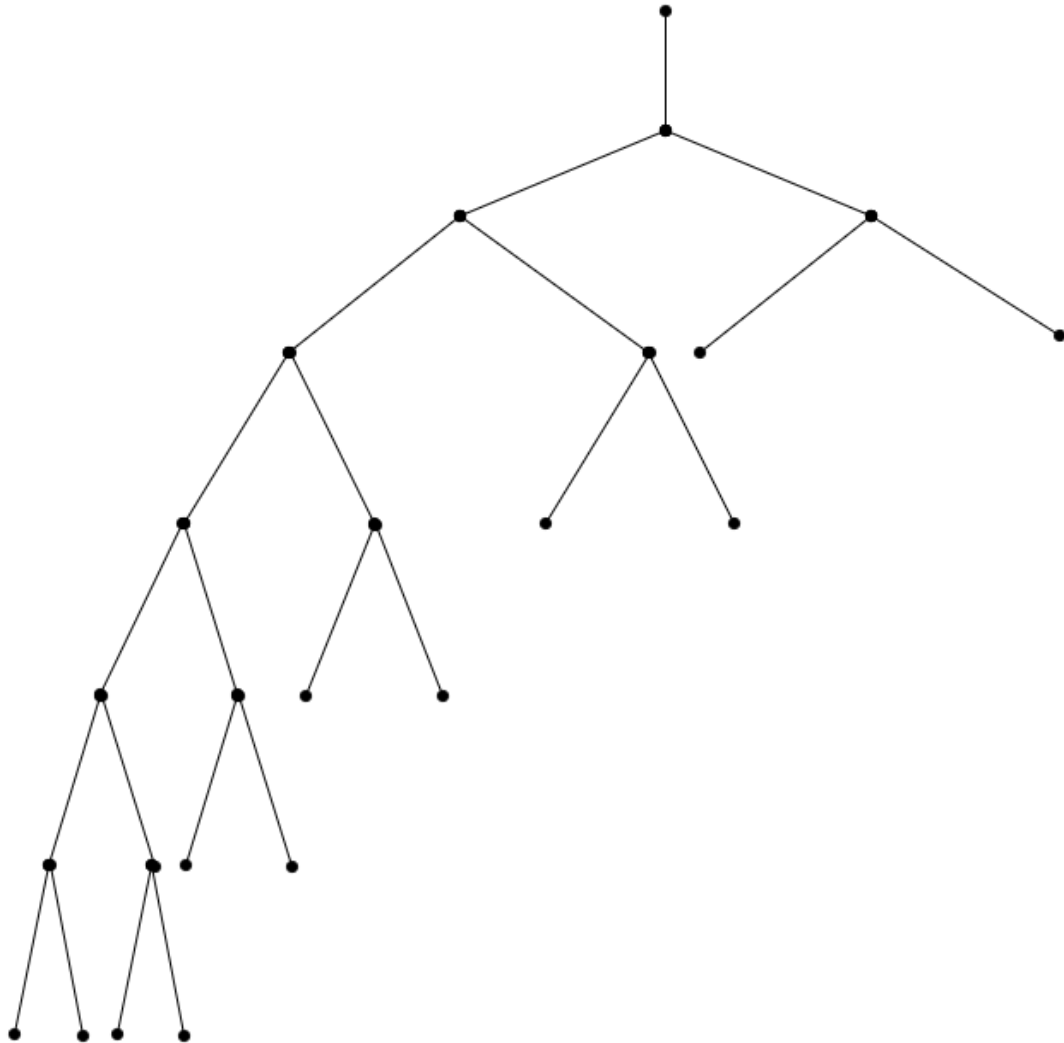
$$\begin{aligned} P(X_{36}) &= \sum_{X_1} \sum_{X_2} \dots \sum_{X_{35}} \sum_{X_{37}} \dots \sum_{X_{1000}} P(X_1)P(X_2|X_1), \dots, P(X_{36}|X_{35})P(X_{37}|X_{36}), \dots P(X_{1000}|X_{999}) \\ &= \sum_{X_1} P(X_1) \sum_{X_2} P(X_2|X_1) \dots \sum_{X_{35}} P(X_{35}|X_{34})P(X_{36}|X_{35}) \sum_{X_{37}} P(X_{37}|X_{36}) \dots \sum_{X_{1000}} P(X_{1000}|X_{999}) \end{aligned}$$

**Inference (complexity without CI: 2^N)

Conditional Independence in a Bayesian net
reduces the computational complexities in computing the query.



**Inference (complexity with CI: $O(N)$)



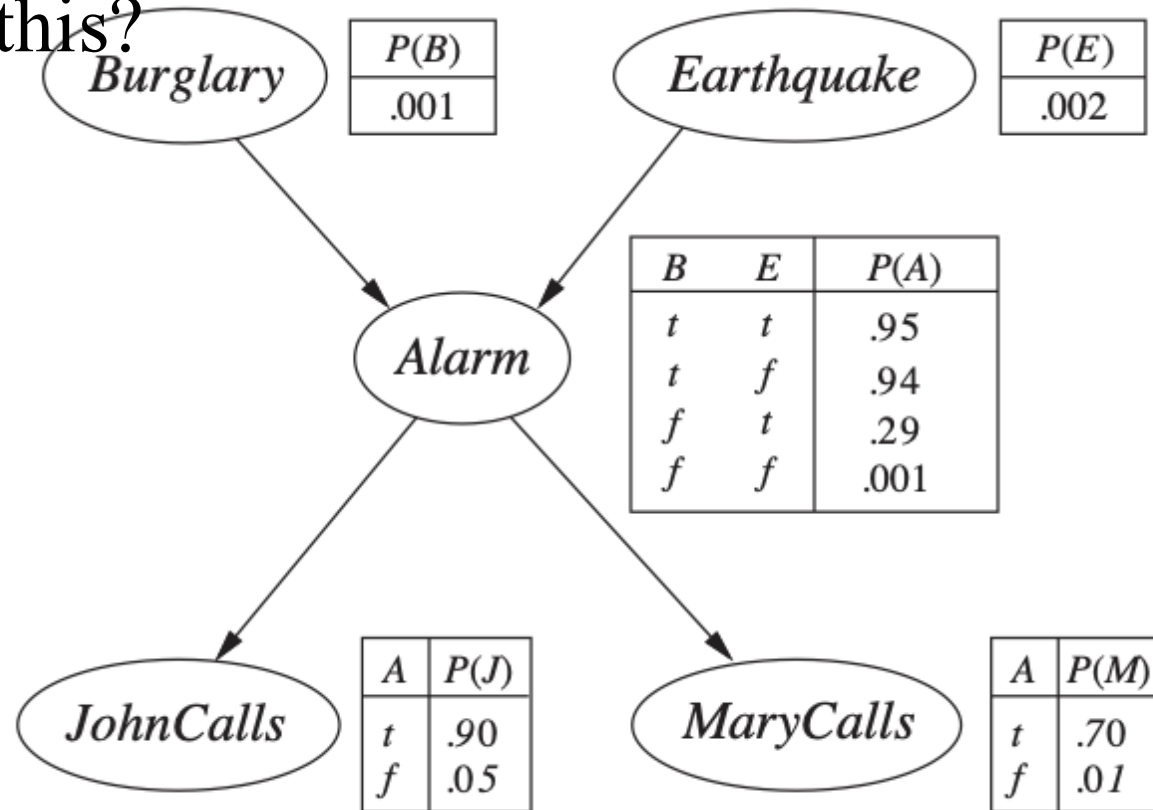
without CI: $\sim O(2^N)$ operations

with CI (Markov Chain Structure): $\sim O(N)$

[3] Inference (computing posterior prob example)

Q: $P[B = + \text{ and } M = +]$?

how can we compute this?



[4] Inference (computing posterior prob example)

$$P[B|J = +, M = +] = \frac{P[B, J = +, M = +]}{P[J = +, M = +]} = \alpha \cdot P[B, J = +, M = +]$$
$$\alpha \cdot P[B, J = +, M = +] = \alpha \cdot \left[\begin{array}{l} P[B = +, J = +, M = +] \\ P[B = -, J = +, M = +] \end{array} \right]$$

Only we need to compute is $P[B, J = +, M = +]$.

$$\alpha (P[B = +, J = +, M = +] + P[B = -, J = +, M = +]) = 1$$

[5] Inference (computing posterior prob example)

How to compute

$$P[B, J = +, M = +] = \left| \frac{P[B = +, J = +, M = +]}{P[B = -, J = +, M = +]} \right|?$$

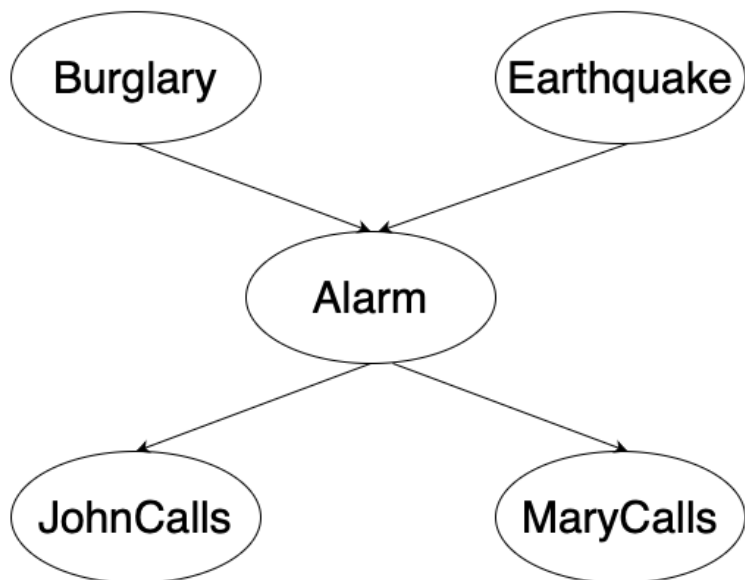
[6] Inference (computing posterior prob example)

By marginalization,

$$P[B, J = +, M = +] = \sum_E \sum_A P[B, E, A, J = +, M = +]$$

[7] Inference (computing posterior prob example)

$$\begin{aligned} P[B, J = +, M = +] &= \sum_E \sum_A P[B, E, A, J = +, M = +] \\ &= P[B] \sum_E P[E] \sum_A P[A|B, E] P[J = +|A] P[M = +|A] \\ &\quad \underbrace{f_1(B)} \quad \underbrace{f_2(E)} \quad \underbrace{f_3(A, B, E)} \quad \underbrace{f_4(A)} \quad \underbrace{f_5(A)} \end{aligned}$$

We need to compute from right to left direction.

[8] Inference (computing posterior prob example)

$$\begin{aligned} P[B, J = +, M = +] &= \sum_E \sum_A P[B, E, A, J = +, M = +] \\ &= P[B] \sum_E P[E] \sum_A \underbrace{P[A|B, E]}_{f_3(A, B, E)} \underbrace{P[J = +|A]}_{f_4(A)} \underbrace{P[M = +|A]}_{f_5(A)} \\ &\quad \underbrace{f_1(B) \quad f_2(E) \quad f_3(A, B, E) \quad f_4(A) \quad f_5(A)}_{f_6(A)} \end{aligned}$$

$$f_6[A] = \begin{bmatrix} f_6[A = +] \\ f_6[A = -] \end{bmatrix} = \begin{bmatrix} P[J = +|A = +] \\ P[J = +|A = -] \end{bmatrix} \begin{bmatrix} P[M = +|A = +] \\ P[M = +|A = -] \end{bmatrix}$$

[9] Inference (computing posterior prob example)

$$\begin{aligned}
 P[B, J = +, M = +] &= \sum_E \sum_A P[B, E, A, J = +, M = +] \\
 &= \underbrace{P[B]}_{f_1(B)} \sum_E \underbrace{P[E]}_{f_2(E)} \sum_A \underbrace{P[A|B, E]}_{f_3(A, B, E)} \underbrace{P[J = +|A]}_{f_4(A)} \underbrace{P[M = +|A]}_{f_5(A)} \\
 &\quad \underbrace{\hspace{10em}}_{f_6(A)} \\
 &\quad \text{---} f_7(BE) = \sum_A f_3(A, B, E) f_6(A)
 \end{aligned}$$

$$\begin{aligned}
 f_7(BE) &= \sum_A P[A|BE] f_6[A] \\
 &= f_6(A+) \begin{bmatrix} P[A+|B+E+] & P[A+|B+E-] \\ P[A+|B-E+] & P[A+|B-E-] \end{bmatrix} + f_6(A-) \begin{bmatrix} P[A-|B+E+] & P[A-|B+E-] \\ P[A-|B-E+] & P[A-|B-E-] \end{bmatrix}
 \end{aligned}$$

[10] Inference (computing posterior prob example)

$$\begin{aligned} P[B, J = +, M = +] &= \sum_E \sum_A P[B, E, A, J = +, M = +] \\ &= \underbrace{P[B]}_{f_1(B)} \sum_E \underbrace{P[E]}_{f_2(E)} \sum_A \underbrace{P[A|B, E]}_{f_3(A, B, E)} \underbrace{P[J = +|A]}_{f_4(A)} \underbrace{P[M = +|A]}_{f_5(A)} \\ &\quad \underbrace{\hspace{10em}}_{f_7(BE)} \\ &\quad \underbrace{\hspace{10em}}_{f_8(B)} \end{aligned}$$
$$f_8(B) = \sum_E f_2(E) f_7(BE)$$

$$f_8(B) = P(E+)f_7(B, E+) + P(E-)f_7(B, E-)$$

[11] Inference (computing posterior prob example)

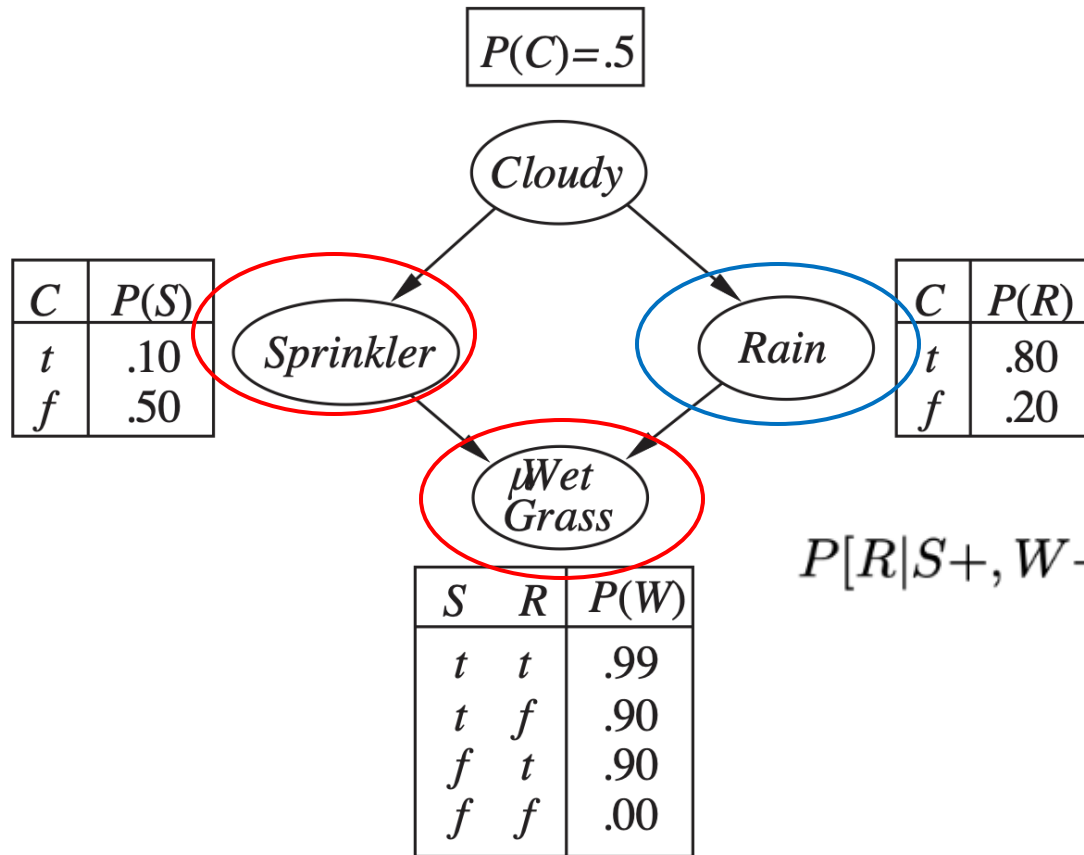
$$\begin{aligned}
 P[B, J = +, M = +] &= \sum_E \sum_A P[B, E, A, J = +, M = +] \\
 &= \underbrace{P[B]}_{f_1(B)} \sum_E \underbrace{P[E]}_{f_2(E)} \sum_A \underbrace{P[A|B, E]}_{f_3(A, B, E)} \underbrace{P[J = +|A]}_{f_4(A)} \underbrace{P[M = +|A]}_{f_5(A)} \\
 &\quad \underbrace{\qquad \qquad \qquad}_{f_8(B) = \sum_E f_2(E) f_7(BE)} \\
 &\quad \underbrace{\qquad \qquad \qquad}_{f_9(B) = P[B] f_8(B)} \\
 P[B, J = +, M = +] &= f_1(B) f_7(B) \\
 P[B|J = +, M = +] &= \alpha f_1(B) f_7(B) \\
 \alpha f_1(B = +) f_7(B = +) + \alpha f_1(B = -) f_7(B = -) &= 1
 \end{aligned}$$

❖ Approximate Inference

[1] Inference (Markov Chain Monte Carlo Simulation)

Q: $P[R \mid S: +, W: +]$?

- [Method 1: Variable Elimination]



$$\begin{aligned} P[R|S+, W+] &= \alpha \sum_C P[R, S+, W+, C] \\ &= \alpha \sum_C P(C) \cdot P(S+|C) \cdot P(R|C) \cdot P(W+|S+R) \\ &= \alpha P(W+|S+R) \sum_C P(C) \cdot P(S+|C) \cdot P(R|C) \end{aligned}$$

Figure 14.12 From the book “AI: A Modern Approach”

[2] Inference (Markov Chain Monte Carlo Simulation)

Q: $P[R : +, W : +]$?

■ [Method 2: Sampling]

$\#samples(Rain: +, Sprinkler: +, WetGrass: +)$

$\# samples (Sprinkler: +, WetGrass: +)$

# data	Rain	Cloudy	Sprinkler	WetGrass
1	+	+	+	+
2	-	-	+	+
3	+	+	+	+
4	+	-	+	+
5	-	+	+	+
6	-	-	+	+
...	-	-	+	+

Fig

[3] Inference (Markov Chain Monte Carlo Simulation)

Q: $P[R : +, W : +]$?

- (1) Gibbs sampling for Bayesian network starts with an arbitrary state
(with the evidence variable fixed at their observed values)

Rain: + Cloud: - **Sprinkler: +, WetGrass: +**

suppose the initial state is this. + fixed as evidence

[4] Inference (Markov Chain Monte Carlo Simulation)

Q: $P[R : +, W : +]$?

(1) sampling for Bayesian network starts with an arbitrary state
(with the evidence variable fixed at their observed values)

Rain: + Cloud: - **Sprinkler: +, WetGrass: +**

suppose the initial state is this.

+ fixed as evidence

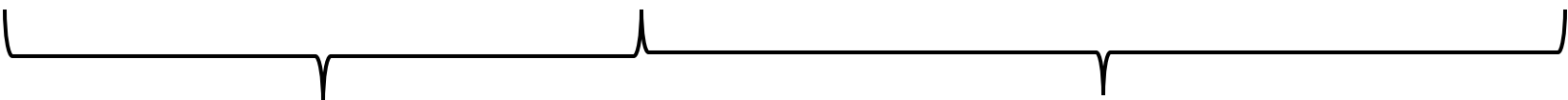
(2) Randomly sample for non evidence variable.
Rain or Cloud (in an arbitrary order)

[5] Inference (Markov Chain Monte Carlo Simulation)

Q: $P[R: +, W: +]$?

(1) sampling for Bayesian network starts with an arbitrary state
(with the evidence variable fixed at their observed values)

Rain: + Cloud: - **Sprinkler: +, WetGrass: +**


+ suppose the initial state is this. + fixed as evidence

(2) Randomly sample for non evidence variable.
Rain or Cloud (in an arbitrary order)

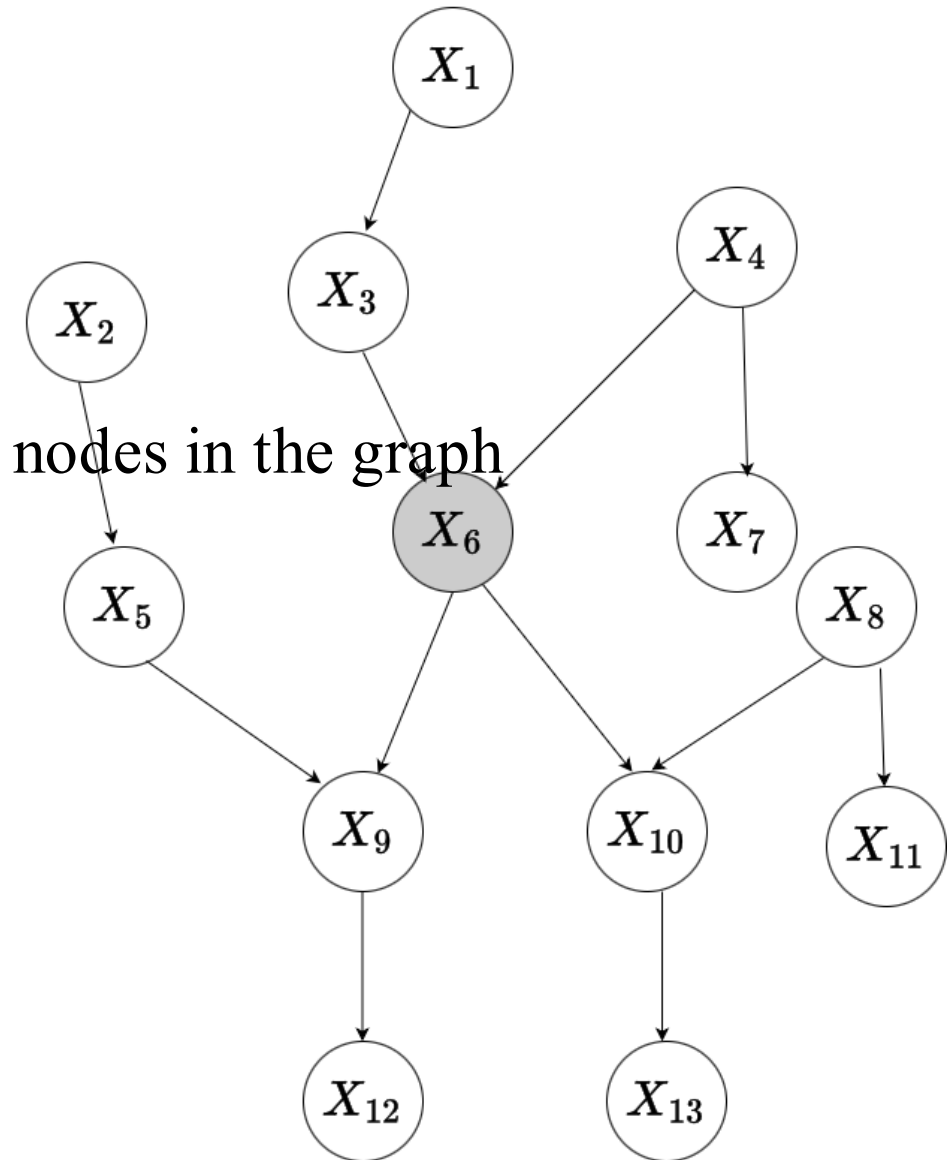
(3) **Cloud is sampled by $P[\text{Cloud} | \text{Sprinkler: + and Rain: +}]$**
given the current value in the Markov Blanket Variables of “Cloud”.

Q: what is Markov Blanket?

** Markov Blanket

Markov Blanket of a Variable

The variable is conditionally independent of every other nodes in the graph given its Markov blanket.



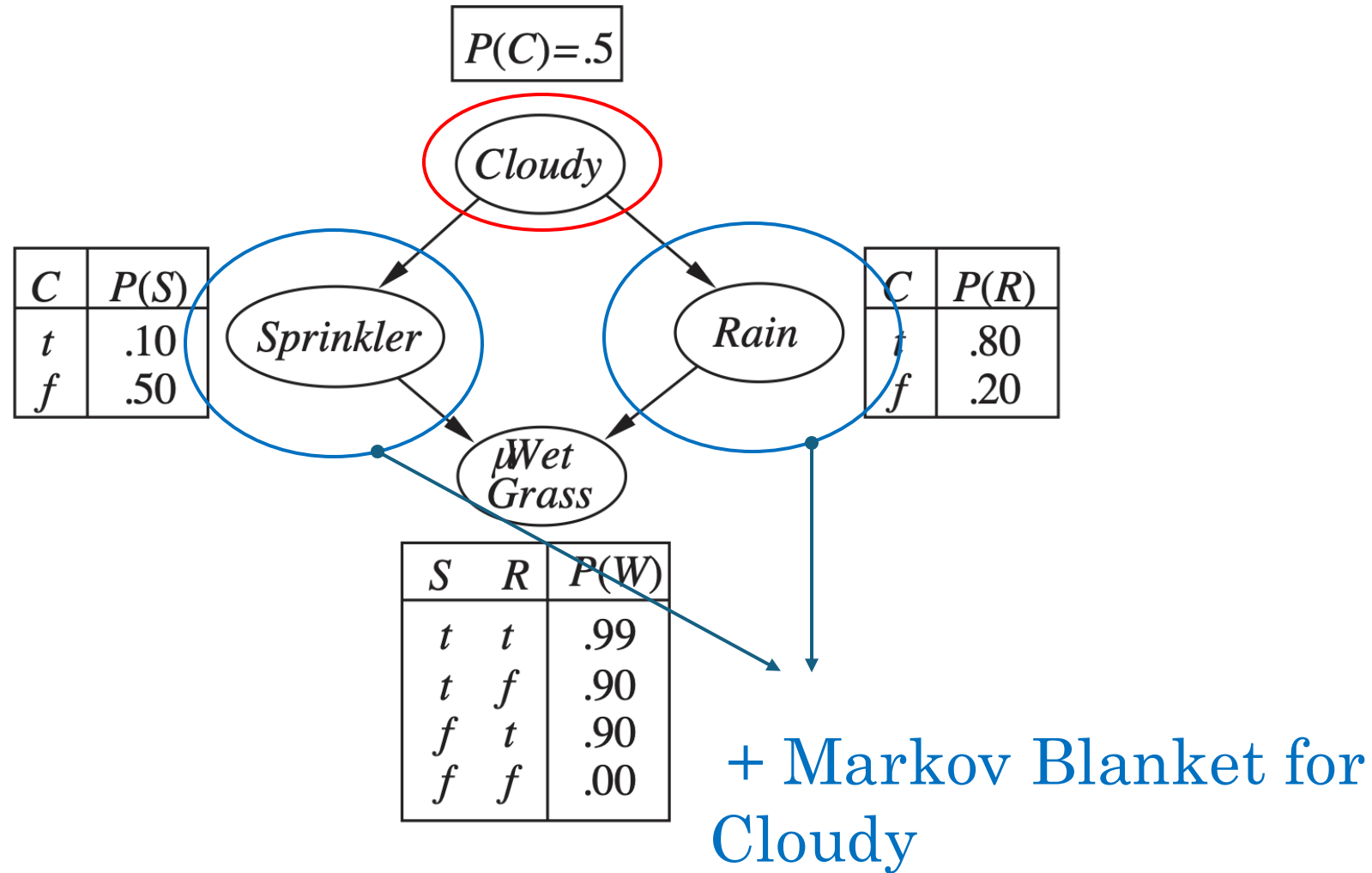
** Markov Blanket

**How to compute $P[\text{Target} \mid \text{Markov Blanket}]$?

Given Markov Blanket, the target variable is independent from other R.Vs, so we can imagine a new graph only with the target and blanket variables.

$$\begin{aligned} P[\text{Target} \mid \text{Markov Blanket}] &= \alpha P[\text{target}, \text{Markov Blanket}] \\ &= \alpha P[\text{target}, \text{parent}, \text{children}, \text{coparent}] \\ &= \alpha P[\text{parent}] P[\text{coparent}] P[\text{target} \mid \text{parent}] P[\text{children} \mid \text{coparents and target}] \\ &= \beta P[\text{target} \mid \text{parent}] P[\text{children} \mid \text{coparents and target}] \end{aligned}$$

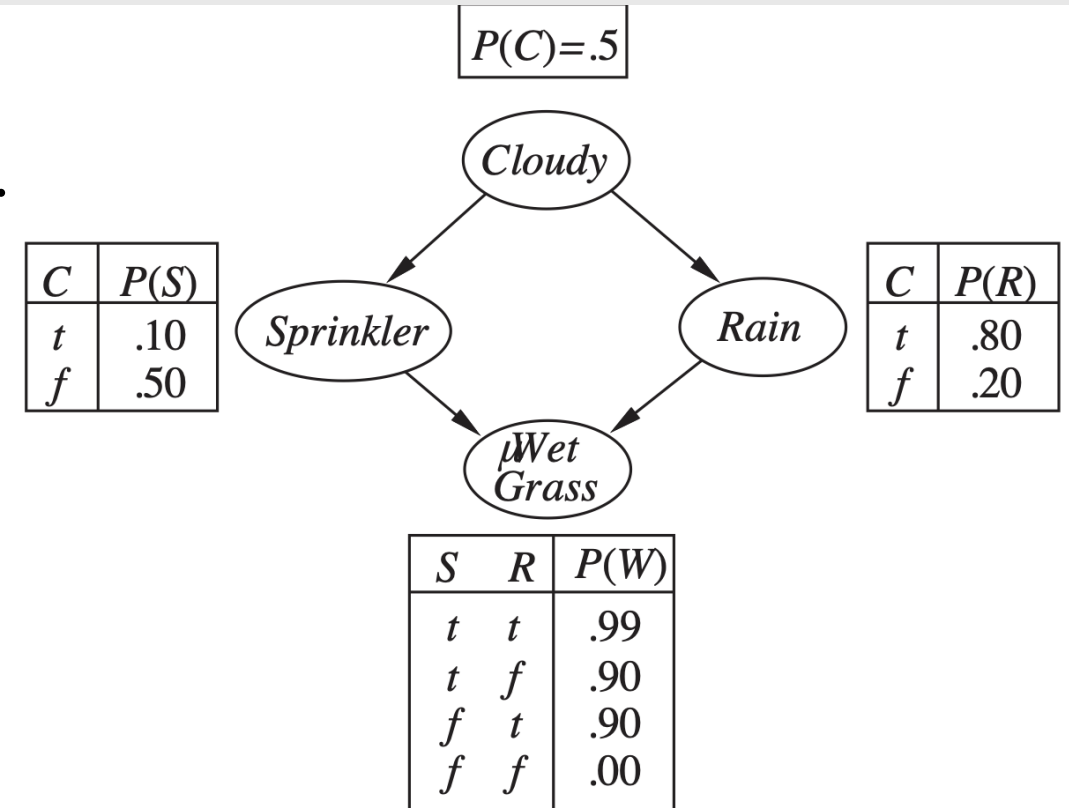
[6] Inference (Markov Chain Monte Carlo Simulation)



Q: What is the Markov Blanket for Rain?

[7] Inference (Markov Chain Monte Carlo Simulation)

Example of Computing Posterior,
given Markov Blanket Variables as Evidence.



$P[\text{Cloud} \mid \text{Sprinkler: + \& Rain: +}]?$

$$\propto \frac{P[C +]P[\text{Sprinkler}+ \mid C+] P[\text{Rain}+ \mid C+]}{P[C -]P[\text{Sprinkler}+ \mid C-] P[\text{Rain}+ \mid C-]} = \propto \begin{bmatrix} 0.04 \\ 0.05 \end{bmatrix} \approx \begin{bmatrix} 0.44 \\ 0.56 \end{bmatrix}$$

Learning a Bayesian Net

- we will focus on the case where a structure is given.

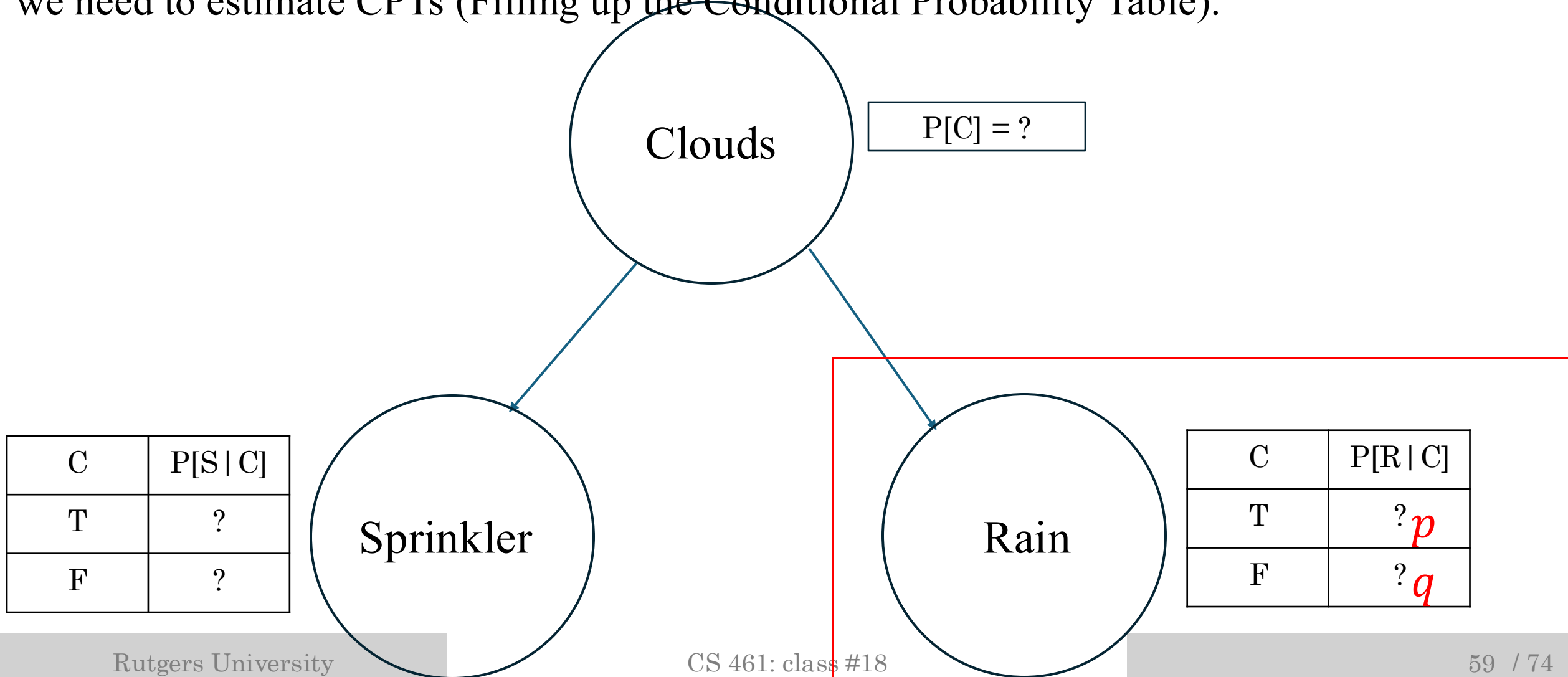
[1] Learning Bayesian Tree

If you are interested in learning a structure,
Chow-Liu Algorithm finds an optimal T (first / second order dependence)
 $T = \operatorname{argmin}_T KL(P || T)$, where P is a true distribution.

Q: what is the true distribution? How can we compute this?

[2] Learning Bayesian Tree (complete data observation)

When a structure is known and data is fully observed, we need to estimate CPTs (Filling up the Conditional Probability Table).



[3] Learning Bayesian Tree (complete data observation)

MLE estimation for p :
when data is fully observed.

$$p(D) = \prod_{n=1}^N P(x_n)$$

$$\log P(D) = \sum_{n=1}^N \log P(x_n)$$

$$= \sum_{n=1}^N \log P[C(x_n)] + \log P[S(x_n)|C(x_n)] + \log P[R(x_n)|C(x_n)]$$

$$\frac{\partial \log P(D)}{\partial p} = N_p \cdot 1/P - N_n \cdot 1/(1 - P) = 0$$

$$P = \frac{N_p}{N_p + N_n}$$

the four possible cases of R and C
: $p, 1 - p, q, 1 - q$

- N_p : # samples cloud + & rain +
- N_n : # samples cloud + & rain -

This is just a sample mean (Bernoulli R.V)

[4] Learning Bayesian Tree (partial data observation)

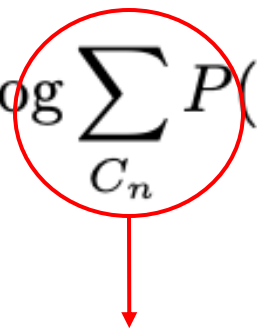
what if data is partially observed?

suppose we forgot to collect **cloud** information.

can we still estimate *p or q* : $P[\text{rain} | \text{cloud} +]$ or $P[\text{rain} | \text{cloud} -]$?

[5] Learning Bayesian Tree (partial data observation)

- MLE when data is partially observed.

$$\log P(D|\theta, p, q) = \sum_{n=1}^N \log \sum_{C_n} P(S_n, R_n, C_n|\theta, p, q)$$


marginalization over unobserved variables

- MLE when data is fully observed (complete MLE)

$$\log P(D|\theta, p, q) = \sum_{n=1}^N \log P(S_n, R_n, C_n|\theta, p, q)$$

Q: which one is easier in computing MLE ?

[6] Learning Bayesian Tree (partial data observation)

$$\begin{aligned}\log P(D|\theta, p, q) &= \sum_{n=1}^N \log P(S_n, R_n, C_n|\theta, p, q) \\ &= \sum_{n=1}^N \log P(C_n) + \log P(S_n|C_n) + \log P(R_n|C_n)\end{aligned}$$

$$\log P(D|\theta, p, q) = \sum_{n=1}^N \log \sum_{C_n} P(S_n, R_n, C_n|\theta, p, q)$$

$$\begin{aligned}&= \sum_{n=1}^N \log(P(C_n+) \cdot P(S_n|C_n+) \cdot P(R_n|C_n+) \\ &\quad + P(C_n-) \cdot P(S_n|C_n-) \cdot P(R_n|C_n-))\end{aligned}$$

hard to optimize (parameters are interdependent)!

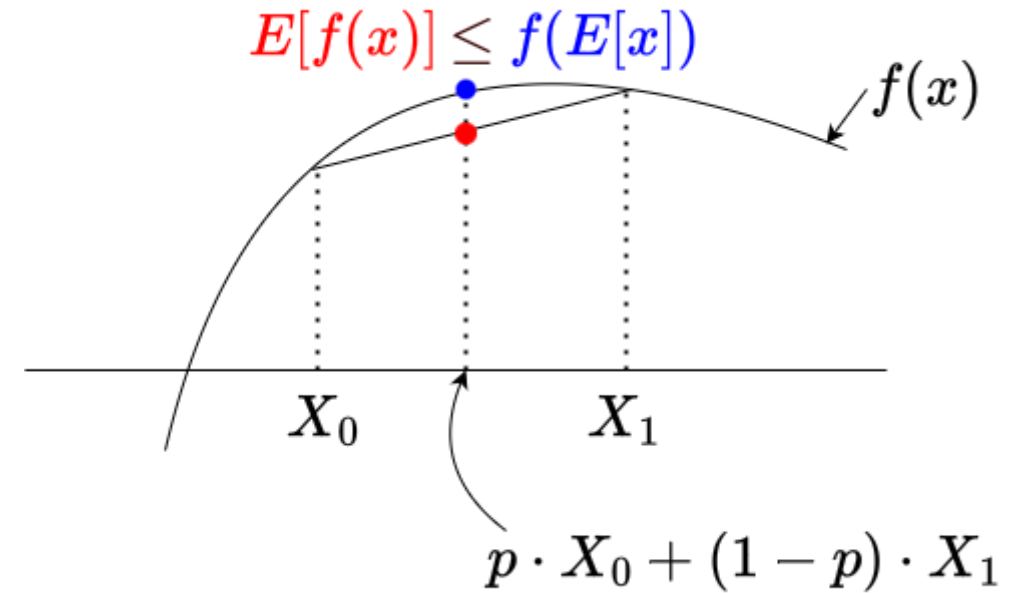
[7] Learning Bayesian Tree (partial data observation)

It's hard to get a closed form to compute MLE,
when there are missing variables.

($\log \sum$ is not tractable for MLE; can we change it to $\sum \log$ like the complete case?)

[8] Learning Bayesian Tree (partial data observation)

Jensen's Inequality: (both expectation are finite)
 $E[\log X] \leq \log E[X]$



[9] Learning Bayesian Tree (partial data observation)

$$\begin{aligned}\log P(D|\theta) &= \sum_{n=1}^N \log \sum_{C_n} P(S_n, R_n, C_n|\theta) \\&= \sum_{n=1}^N \log \sum_{C_n} \frac{P(S_n, R_n, C_n|\theta)q(C_n)}{q(C_n)} \\&= \sum_{n=1}^N \log E\left[\frac{P(S_n, R_n, C_n|\theta)}{q(C_n)}\right] \\&\geq \sum_{n=1}^N E\left[\log\left(\frac{P(S_n, R_n, C_n|\theta)}{q(C_n)}\right)\right] \\&\geq \sum_{n=1}^N \underbrace{E[\log P(S_n, R_n, C_n|\theta)] + H(q(C_n))}\end{aligned}$$

- hard to optimize!

- by Jensen's Inequality

- easy to optimize!

expectation over

$q(C_n)$: any arbitrary density for C_n

gives a lower bound.

Q: how can we make the lower bound tight?

[10] Learning Bayesian Tree (partial data observation)

Q: What does the expectation of likelihood mean?

Likelihood is not a probability. It is a function of the parameters.

In the example below, the likelihood with one point data x_i is a function of $L(\mu, \sigma)$.

$$\mathcal{N}(x_i|\mu, \sigma^2) = \frac{1}{\sqrt{(2\pi\sigma^2)}} \exp\left\{\frac{-1}{2\sigma^2}(x_i - \mu)^2\right\}$$

What if x_i is unobserved and instead we know the density $x_i \sim p(x_i)$?
we can measure the expectation treating μ *and* σ as constants.

$$E[f(X_i|\mu, \sigma)] = \sum_{x_i} f(x_i|\mu, \sigma)p(x_i)$$

[11] Learning Bayesian Tree (partial data observation)

Q: what density makes the lower bound tight?

$$\begin{aligned} & \sum_{n=1}^N E[\log P(S_n, R_n, C_n | \theta) + H(q(C_n))] \\ &= \sum_{n=1}^N \sum_{C_n} q(C_n) \log \left(\frac{P(S_n, R_n, C_n | \theta)}{q(C_n)} \right) \\ &= \sum_{n=1}^N \sum_{C_n} q(C_n) \log \left(\frac{P(C_n | S_n, R_n, \theta) P(S_n, R_n | \theta)}{q(C_n)} \right) \\ &= \sum_{n=1}^N \left(\sum_{C_n} q(C_n) \log \frac{P(C_n | S_n, R_n, \theta)}{q(C_n)} + \sum_{C_n} q(C_n) \log P(S_n, R_n | \theta) \right) \\ &= \sum_{n=1}^N -KL(q(C_n) || P(C_n | S_n, R_n, \theta)) + \log P(S_n, R_n | \theta) \end{aligned}$$

when $q(C_i) = P(C_i | S_i, R_i, \theta, p, q)$

the lower bound becomes equal to $\log P(S_i, R_i | \theta, p, q)$

[12] Learning Bayesian Tree (partial data observation)

$$\begin{aligned}\log P(D|\theta) &= \sum_{n=1}^N \log \sum_{C_n} P(S_n, R_n, C_n|\theta) \\ &= \sum_{n=1}^N \log \sum_{C_n} \frac{P(S_n, R_n, C_n|\theta)q(C_n)}{q(C_n)} \\ &= \sum_{n=1}^N \log E\left[\frac{P(S_n, R_n, C_n|\theta)}{q(C_n)}\right] \\ &\geq \sum_{n=1}^N E\left[\log\left(\frac{P(S_n, R_n, C_n|\theta)}{q(C_n)}\right)\right] \\ &\geq \sum_{n=1}^N E[\log P(S_n, R_n, C_n|\theta)] + H(q(C_n))\end{aligned}$$

expectation over $q(C_i) = P(C_i | S_i, R_i, \theta, p, q)$

what is the next ?

we do maximization over θ, p, q . why?

Expectation Maximization:
finding maximum likelihood when data is partially observed.

[1] EM algorithm (cloud, sprinkler, rain example)

1. start with arbitrary parameters: θ, p, q
2. compute $P(C_n | S_n, R_n, \theta, p, q)$ for $\forall n$
3. **E step**: compute

$$\sum_{n=1}^N E[\log P(S_n, R_n, C_n | \theta, p, q)]$$

4. **M step**: Update $\theta, p, q = \operatorname{argmax}_{\theta, p, q} \sum_{n=1}^N E[\log P(S_n, R_n, C_n | \theta, p, q)]$
5. go back to step 2.

[2] EM algorithm (cloud, sprinkler, rain example)

1. start with arbitrary parameters: θ, p, q

2. compute $P(C_i | S_i, R_i, \theta, p, q)$ for $\forall i$

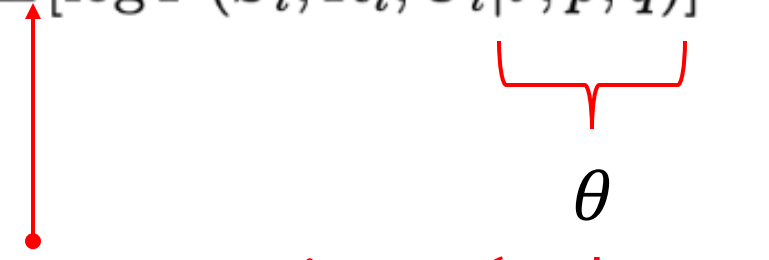
3. E step: compute $\sum_{n=1}^N E[\log P(S_i, R_i, C_i | \theta, p, q)] = Q(\theta, \theta^t)$

$\downarrow$
Auxiliary Function

4. S step: Update $\theta, p, q = \operatorname{argmax}_{\theta, p, q} \sum_{n=1}^N E[\log P(S_i, R_i, C_i | \theta, p, q)]$

5. go back to step 2.

$$\text{Auxiliary Function } Q(\theta, \theta^t) = \sum_{n=1}^N E[\log P(S_i, R_i, C_i | \theta, p, q)]$$



expectation $P(C_i | S_i, R_i, \theta^t)$

- $Q(\theta^t, \theta^t) = \sum_i \log P[S_i, R_i | \theta^t]$
- $Q(\theta^{t+1}, \theta^t) \geq Q(\theta^t, \theta^t)$ by **Maximization** EM monotonically increases the observed data log likelihood!
- $Q(\theta^{t+1}, \theta^{t+1}) \geq Q(\theta^{t+1}, \theta^t)$ by **lower bound**
- $Q(\theta^{t+1}, \theta^{t+1}) = \sum_i \log P[S_i, R_i | \theta^{t+1}]$ by **Expectation**
- $\sum_i \log P[S_i, R_i | \theta^t] \leq Q(\theta^{t+1}, \theta^t) \leq Q(\theta^{t+1}, \theta^{t+1}) = \sum_i \log P[S_i, R_i | \theta^{t+1}]$

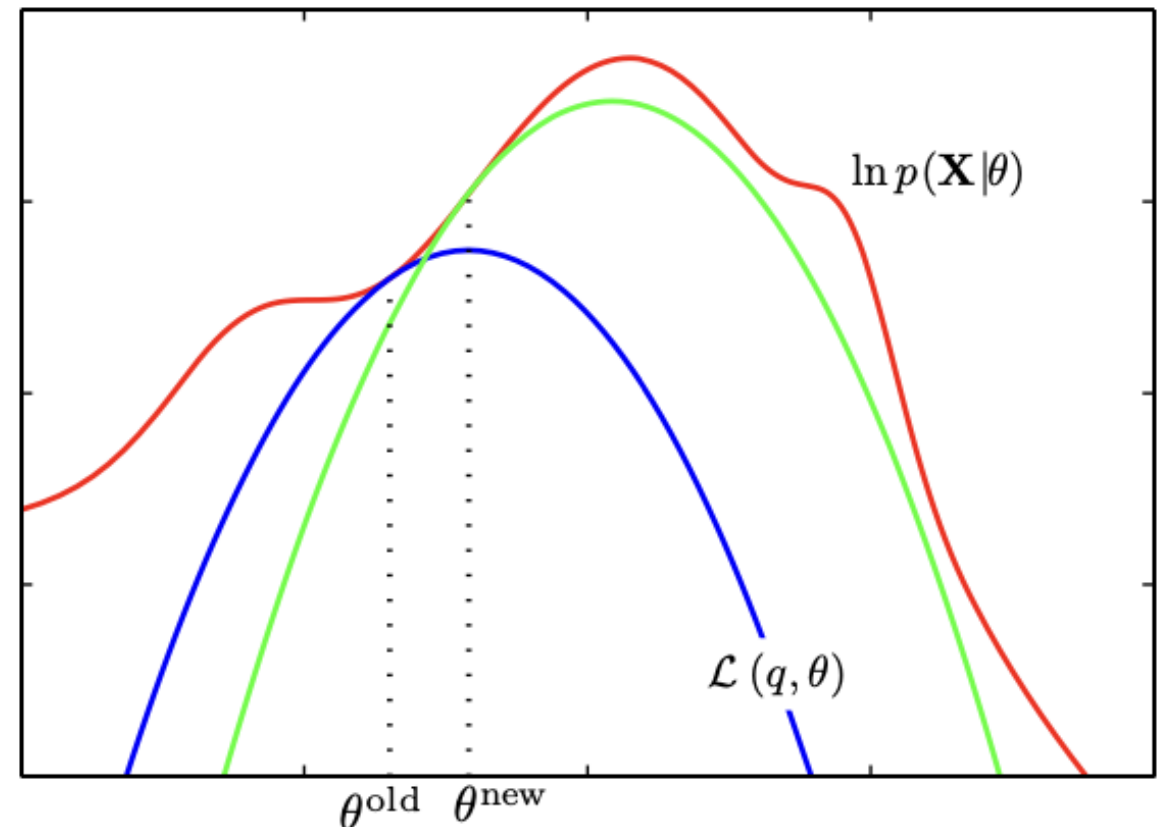
[3] EM algorithm (original likelihood vs. auxiliary function)

EM algorithm finds a local minimum.

$\log P(\mathbf{X}|\theta)$ is often non – convex or non – concave.

Figure 9.14 The EM algorithm involves alternately computing a lower bound on the log likelihood for the current parameter values and then maximizing this bound to obtain the new parameter values. See the text for a full discussion.

From textbook Bishop



- Comparison between EM and gradient based methods